

ASWIN K M

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EDUCATION

Lovely Professional University

Bachelor of Technology in Computer Science

Aug. 2022 – May 2026

Punjab, India

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, System Design, Databases, Computer Systems, Computer Networks

PROJECTS

Skill India – Laravel Course Management System | *Laravel, PHP, MySQL*

GitHub

- Collaborated within a team to build a dual-module course management platform using **Laravel** and PHP, architecting independent admin and learner-facing applications that share a centralized **MySQL** database.
- Implemented role-based authentication, full CRUD operations for course management, and session-based user registration, login, and course enrollment workflows.
- Designed relational database schemas using Laravel migrations and seeders, and built an admin dashboard for real-time course and enrollment analytics.

RideSync – Real-Time Spatial Dispatch Engine | *Go, PostgreSQL, Redis, Docker, AWS*

GitHub

- Architected a high-performance real-time ride-hailing backend in **Go**, taking full ownership of the Low-Level Design (LLD) and system architecture to dynamically match users in **<2ms**.
- Designed and optimized a highly concurrent full-duplex WebSocket pipeline using Go channels, sustaining **10,000+** simultaneous telemetry streams without data race conditions.
- Reduced SQL database I/O overhead by **99.8%** by building asynchronous background workers for batch processing, integrating seamlessly into a cloud-native deployment.

SeatForge – Distributed Seat Reservation System | *Spring Boot, PostgreSQL, Kafka, Kubernetes*

GitHub

- Built and maintained a distributed system for seat reservations, guaranteeing strict data consistency and fault tolerance in high-concurrency transactions to prevent double-booking.
- Migrated architecture to a microservices architecture and deployed via Docker and Kubernetes behind an Nginx load balancer, achieving **3× throughput** and sustaining **5,000 RPS**.
- Optimized backend performance by integrating Redis caching and a Kafka messaging queue, reducing database load by **65%** and eliminating N+1 query bottlenecks.

TECHNICAL SKILLS

Languages : PHP, Go (Golang), Java, Python, C/C++, JavaScript, SQL, NoSQL

Backend Services : Laravel, RESTful APIs, MVC Architecture, Spring Boot, FastAPI, Node.js

Architecture & Design : Low-Level Design (LLD), System Design, Microservices Architecture, Distributed Systems

Infrastructure & Cloud : AWS, Docker, Kubernetes, CI/CD, Linux, Monitoring & Logging

Databases & Messaging : MySQL, PostgreSQL, Redis, MongoDB, Kafka, RabbitMQ

ACHIEVEMENTS & CERTIFICATIONS

- **LeetCode:** Solved **400+** problems (161 Easy, 190 Medium, 50 Hard) with an **82.86%** acceptance rate; Global Ranking **1,700**.
- **Standard Membership, Google Developer Program:** Active member utilizing enterprise tools to troubleshoot, debug, and optimize application code.
- **Smart India Hackathon (SIH):** Ranked **Top 10** internally with a **97/100** evaluation score; showcased strong problem-solving skills in a highly collaborative environment.
- **Certifications:** Cloud Computing (NPTEL), Blockchain Specialization (Coursera).